RESEARCH REPORT

Concept Paper: From Flat Composite Intent to Hierarchical Intent Classification (Action + Facet + Slots)

Intent Routing and Retrieval Strategy Redesign

Generated automatically from markdown source

Source: CONCEPT_PAPER_INTENT_V2.md

Concept Paper: From Flat Composite Intent to Hierarchical Intent Classification (Action + Facet + Slots)

Date: 2026-02-20
Project: GenAI GraphRAG (TMT domain)
Scope: Intent routing design for retrieval and search planning

1) Executive Summary

The current intent design is workable for core in-domain questions, but it does not match real query patterns well enough.
The largest issue is that the current labels are cross-product labels (`facet_action`, such as `manufacturer_find`) with only 3 actions (`find/check/count`).
Real traffic includes patterns that are not represented explicitly (for example ID lookup, list-style requests, compare-like requests, and out-of-domain chat).

This paper proposes Hierarchical Intent Classification (HIC): - `action_intent` (single-label): `lookup`, `verify`, `list`, `count`, `compare`, `unknown` - `facet_intents` (multi-label): `manufacturer`, `ingredient`, `nlem`, `formula`, `hierarchy`, `id_lookup` - `slots` (structured extracted values): `drug`, `tmtid`, `manufacturer`, `strength`, etc.

HIC is expected to improve routing quality and retrieval precision/recall for mixed or compositional queries, with a small acceptable latency increase.

2) Current Approach (As-Is)

Current intent dataset and classifier behavior: - Dataset: `src/api/intent_dataset.json` - Labels: 18 (`6 facets x 3 actions`) - Samples: 270 total - Actions represented: `find`, `check`, `count` only - Facets represented: `manufacturer`, `ingredient`, `nlem`, `hierarchy`, `formula`, `general`

Current AQT behavior: - Vector centroid classifier predicts a fine label and target/facet-like class. - Rule-based strategy maps to `retrieve`, `verify`, `count`. - No explicit `list`, `compare`, or `unknown` action class. - No explicit `id_lookup` facet class.

3) Evidence from Real Data

Real query source used for assessment: - `logs/ragas_data.jsonl` - 44 total queries, 25 unique

Observed real query patterns: - Queries with `TMTID` keyword: 2/44 (4.5%) - Trade-name request style (for example "want trade names"): 17/44 (38.6%) - Compare-like phrase: 1/44 (2.3%) - OOD/chat-like greeting: 1/44 (2.3%)

Runtime intent behavior on these 44 queries (current pipeline): - Target distribution: - `GENERAL`: 20 - `MANUFACTURER`: 14 - `NLEM`: 8 - `FORMULA`: 2 - Strategy distribution: - `RETRIEVE`: 33 - `VERIFY`: 8 - `COUNT`: 3

Confidence quality signal: - Margin `< 0.03`: 22/44 queries (50.0%) - Median margin: 0.0292

Interpretation: - Many queries are routed to generic retrieve behavior. - A large portion of traffic is near decision boundaries. - Existing label space is too narrow for real request variety.

4) Experimental Design

Objective: - Compare Flat Composite Intent (FCI) vs Hierarchical Intent Classification (HIC) under the same data and embedding conditions. - Verify not only in-domain performance, but also compositional generalization.

Primary hypotheses: - H1: In easy in-domain splits, FCI and HIC are similar. - H2: In compositional holdout splits, HIC outperforms FCI on action/facet routing quality.

Systems under test: - FCI (Flat Composite Intent, baseline): - Single centroid classifier over fine labels (`facet_action`, 18 classes). - Predicted fine label is back-parsed to action and facet. - HIC (Hierarchical Intent Classification): - Split-head centroid classifiers: - action head - facet head - Final decision is composed from both heads.

Dataset and source: - Source file: `src/api/intent_dataset.json` - Derived HIC reference file: `experiments/intent_benchmarks/intent_dataset_v2_reference.json` - Total samples used in benchmark: 270 - Note: This benchmark uses existing text only (no new text augmentation yet).

Controlled variables: - Same embedding model for both systems (`bge-m3`). - Same train/test partitions for both systems in each experiment. - Same cosine-centroid nearest prediction rule. - Same preprocessing path for labels and records. - Fixed random seed: 42.

Experimental splits: - Split A: Random Stratified Split - Method: split each fine label into train/test with constant ratio. - Purpose: estimate in-domain performance where train/test distributions are similar. - Split B: Compositional Holdout Split - Method: hide selected fine labels from training, evaluate on those hidden labels only. - Holdout labels used: - `manufacturer_check` - `ingredient_count` - `nlem_find` - `hierarchy_check` - `formula_count` - `general_find` - Purpose: test generalization to unseen action+facet combinations. - Meaning of

"hidden": - FCI never sees those exact fine labels. - HIC can still combine learned action and facet patterns from separate heads.

Evaluation metrics: - `action_acc`, `action_f1`: ability to predict what user wants to do. - `facet_acc`, `facet_f1`: ability to predict which information dimension is needed. - `joint_acc`: strict score, requires both action and facet to be correct on the same sample. - Inference latency: - measured on prediction step only (embedding already computed), - averaged over repeated loops.

Interpretation protocol: - Random split results indicate "fit to seen pattern families". - Compositional holdout results indicate "robustness to unseen composition". - For production-oriented routing, compositional metrics have higher decision weight.

Limitations of current experiment design: - No slot extraction evaluation yet (`tmtid`, manufacturer span, strength span, etc.). - No end-to-end retrieval quality metrics in this benchmark (`P@k`, `R@k`, answer faithfulness). - Single-seed reporting (future work should add repeated seeds and confidence intervals).

5) Experimental Benchmark Evidence

Experiment script: - `experiments/intent_benchmarks/benchmark_intent_v2_vs_legacy.py` Result file: - `experiments/intent_benchmarks/results/benchmark_intent_v2_vs_legacy_20260220_134337.json`

5.1 Random stratified split (in-domain style)

Full metrics table:

Model	action_acc	action_f1	facet_acc	facet_f1	joint_acc	latency_ms
FCI	0.8667	0.8647	0.8333	0.8320	0.7333	0.2077
HIC	0.8667	0.8647	0.8333	0.8343	0.7000	0.2843
Delta (HIC - FCI)	+0.0000	+0.0000	+0.0000	+0.0023	-0.0333	+0.0765

Support and class scope:

Item	Value
test_size	90

Item	Value
FCI class count (fine labels)	18
HIC action class count	3
HIC facet class count	6

Reading: - FCI and HIC are similar on top-line action/facet in easy in-domain split. - HIC has slightly lower strict `joint_acc` in this split.

5.2 Compositional holdout split (harder generalization test)

Full metrics table:

Model	action_acc	action_f1	facet_acc	facet_f1	joint_acc	latency_ms
FCI	0.3333	0.3465	0.6111	0.6077	0.0000	0.2007
HIC	0.5333	0.5360	0.6667	0.6727	0.2556	0.3028
Delta (HIC - FCI)	+0.2000	+0.1895	+0.0556	+0.0650	+0.2556	+0.1021

Support and holdout setting:

Item	Value
test_size	90
FCI class count seen in train	12
HIC action class count	3
HIC facet class count	6

Holdout fine labels: - `manufacturer_check` - `ingredient_count` - `nlem_find` - `hierarchy_check` - `formula_count` - `general_find`

Interpretation: - HIC is clearly better when query combinations are less seen or unseen. - This is the expected production-like benefit for evolving query styles.

5.3 Real Query Case Studies (Production-Like Examples)

Real query sample: - Source: `logs/ragas_data.jsonl` - Evaluated unique queries: 25 - Comparison artifact: `experiments/intent_benchmarks/results/real_query_fci_vs_hic_preview.json` - Cases where FCI and HIC disagree: 6/25

Representative examples:

Real query	Expected intent meaning	FCI output	HIC output	Observation
`พาราอยู่ในบัญชียาหลักใช่หรือไม่`	`verify + nlem`	`lookup + nlem`	`verify + nlem`	HIC aligns better with yes/no verification intent.
`Paracetamol อยู่ในบัญชียาหลักใช่หรือไม่`	`verify + nlem`	`lookup + nlem`	`verify + nlem`	Same pattern in English/Thai mixed form; HIC is more action-correct.
`MACROPHAR ผลิตยาพาราไหม`	`verify + manufacturer`	`lookup + manufacturer`	`verify + manufacturer`	HIC captures verification cue from "ไหม" more reliably.
`paracetamal กี่ ml เป็นอันตราย`	likely safety/unknown (not simple count)	`lookup + formula`	`count + formula`	Both are weak here; action space lacks safety-risk intent.
`สวัสดี ผมชื่ออะไรให้ทาย`	`unknown` (OOD/chit-chat)	`verify + general`	`lookup + general`	Both fail as expected because `unknown` class is not modeled yet.
`ขอ TMTID ของยา cefazolin ...`	`lookup + id_lookup`	`lookup + hierarchy`	`lookup + hierarchy`	Both miss `id_lookup` facet because this facet is absent in current label space.

What these cases show: - HIC improves action semantics on verification-style real queries. - Some production patterns still fail under both methods due to taxonomy gaps, not model weakness alone. - Priority gaps to close in next data design: - `unknown` action - `id_lookup` facet - safety/toxicity style intents (or explicit fallback policy to non-RAG flow)

6) Problem Statement

Why change is necessary: - The current taxonomy couples "what user wants to do" and "which information area is needed" into one label. - Cross-product labels do not scale well and are hard to extend. - Important production intents are not first-class citizens (`list`, `compare`, `unknown`, `id_lookup`). - Routing is overly biased toward generic retrieve path under ambiguity.

7) Proposed Hierarchical Intent Classification (To-Be)

7.1 Intent schema

- `action_intent` (single-label):
- `lookup`, `verify`, `list`, `count`, `compare`, `unknown`
- `facet_intents` (multi-label):
- `manufacturer`, `ingredient`, `nlem`, `formula`, `hierarchy`, `id_lookup`
- `slots` (structured extraction):
- `drug_name`, `tmtid`, `manufacturer`, `strength`, `dose_form`, `nlem_category`, etc.

7.2 Routing logic

- `action_intent` selects operator family:
- `lookup` -> hybrid retrieve
- `verify` -> existence/claim verification flow
- `list` -> list retrieval with filters and limits
- `count` -> aggregate count query
- `compare` -> dual retrieval + diff
- `unknown` -> safe fallback or clarification
- `facet_intents` choose search scope/boost/filter/traversal depth.
- `id_lookup` has highest override priority (exact lookup path first).
- `slots` become deterministic constraints for Cypher and fulltext.

8) Expected Benefits

- Better precision/recall due to clearer routing and constraints.
- Better compositional generalization (supported by holdout benchmark).
- Lower token waste for small local LLM by reducing irrelevant context.
- Better observability and debugging (scores per action/facet + slot source).

9) Trade-offs and Risks

- Slightly higher inference latency (two heads + slot resolver).
- Higher implementation complexity than a single flat classifier.
- Requires re-labeling and data pipeline support for action/facet/slot format.

Mitigation: - Keep fast deterministic pre-checks (`id_lookup`, units, obvious patterns). - Cache embeddings and route decisions. - Add confidence thresholds and fallback policy.

10) Success Criteria for Adoption

Adopt HIC if the following are met: - Action accuracy and facet F1 improve on compositional holdout. - End-to-end retrieval improves on production-like eval set (Precision@k, Recall@k, answer faithfulness). - Token usage per answer decreases for equivalent or better answer quality. - No major latency regression beyond agreed budget.

11) Recommendation

Proceed with HIC design and phased rollout.
Maintain FCI path as fallback during transition, but use HIC as the primary experiment track for production-oriented evaluation.